

Profile-driven generation of standardized, executable FHIR transformation artifacts for interoperable clinical data

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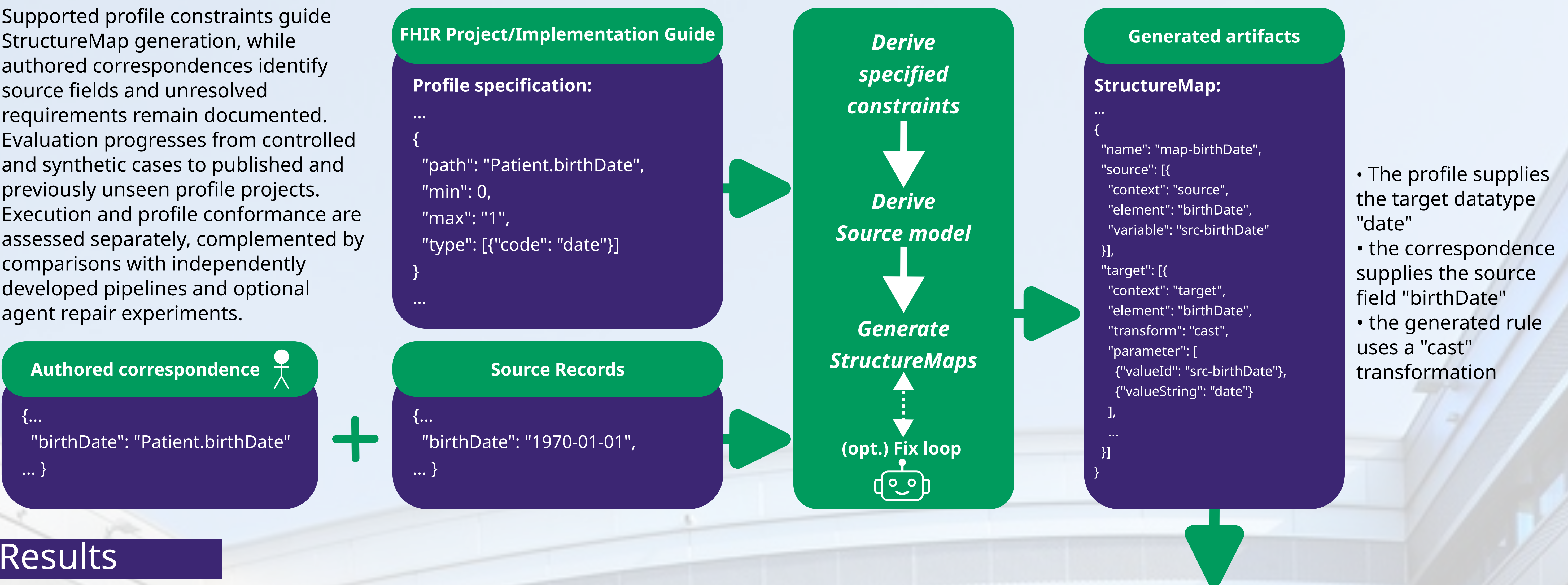
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Introduction

Secondary use of heterogeneous clinical data requires consistent, reusable representations. HL7 Fast Healthcare Interoperability Resources (FHIR) profiles specify constraints on resource structure, cardinalities, and terminology. [1] However, transformation pipelines commonly encode these structural requirements in separately maintained mapping logic, requiring coordinated updates to profiles and implementation. Existing approaches address different aspects of this problem. Redmatch¹ provides a dedicated mapping language for REDCap data, while FUME² and Sinaci et al. [2] support profile-aware transformations through custom mapping languages. Marfoggia et al. [3] demonstrate a modular pipeline using manually authored FHIR StructureMaps. FML2Mirth [4] and MaLaC-HD [5] support execution of existing FHIR Mapping Language (FML) definitions in other environments. Here we show that standardized, executable FHIR StructureMaps can be generated based on profile-derived constraints while documenting unresolved source mappings. We evaluate the approach using controlled, synthetic, and real-world profiles and comparisons with independently developed pipelines. Secondary experiments investigate a feedback-guided agent repair.

Methods

Supported profile constraints guide StructureMap generation, while authored correspondences identify source fields and unresolved requirements remain documented. Evaluation progresses from controlled and synthetic cases to published and previously unseen profile projects. Execution and profile conformance are assessed separately, complemented by comparisons with independently developed pipelines and optional agent repair experiments.



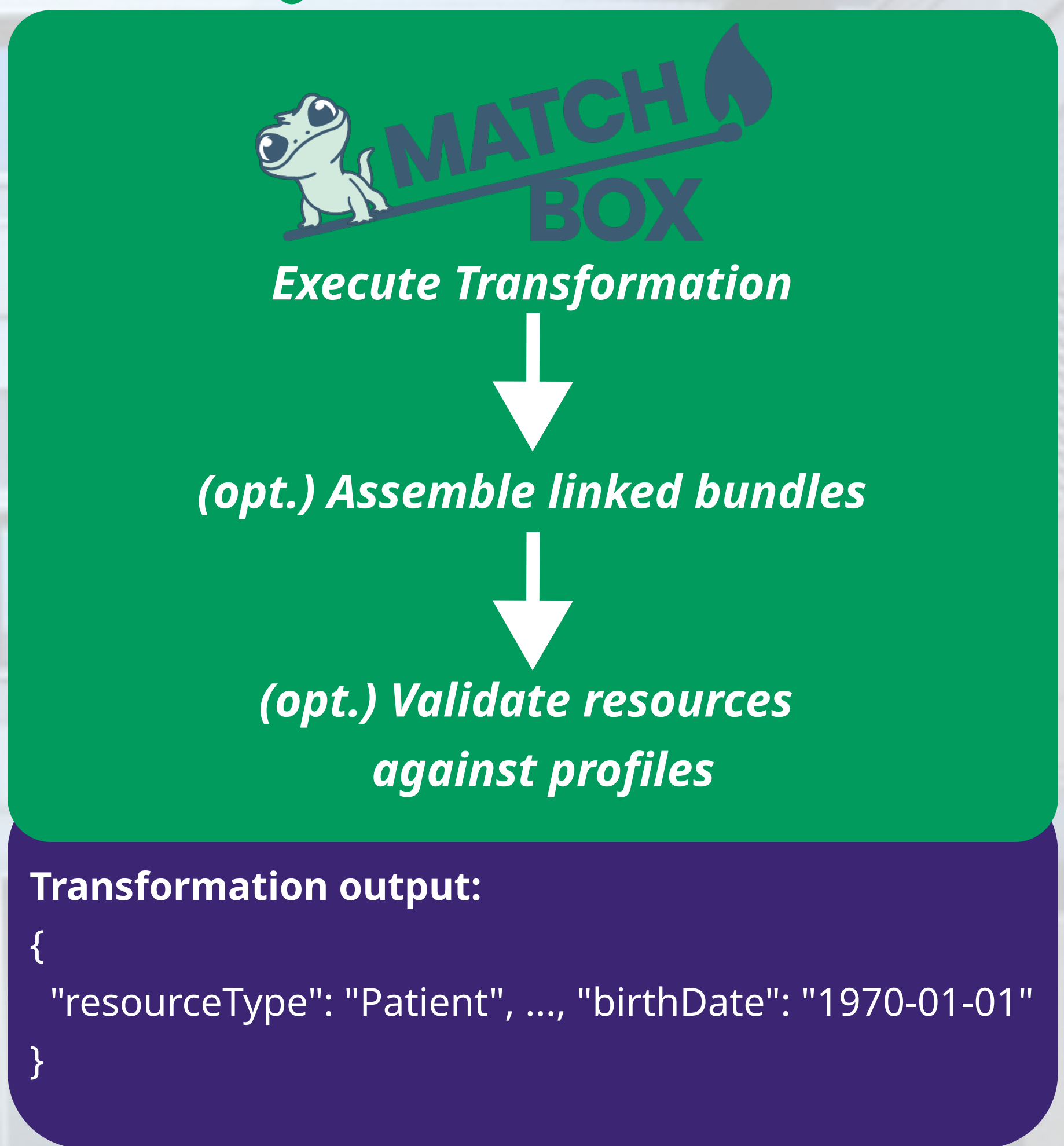
Results

Evaluation across controlled examples, published profile projects, independent pipelines, and optional agent repair. Execution and conformance were assessed separately

Evaluation	Execution/output	Outcome
Controlled: 2 projects	8/8 resources produced	8/8 passed individual validation; 4/4 maps regenerated identically
Synthetic: 6 profiles	6/6 resources produced	5/6 passed individual validation; 6/6 passed after bundle assembly
Real-world: 21 projects	132/132 resources produced	115/132 passed intended-profile validation
Previously unseen: 12 projects	181/183 resources produced	133/180 assessed outputs passed intended-profile validation ^a
Rehabilitation pipeline	6,186 bundles loaded	All five published reference-query results reproduced
Survey pipeline	145 resources; reference: 176	All 22 generated QuestionnaireResponses validated; other output differences remained
Feedback-guided agent repair	Paired comparisons with generated maps	6 accepted patches across all projects ^b

^aOf 183 expected resources, two failed to transform and one produced resource lacked a recorded validation assessment.

^bThree repetitions over the 21 projects accepted 2, 1, and 1 patches. The unseen repair run covered 176 maps across 11 completed projects; one project failed during generation.



Discussion

- Applicability in practice: Generated maps supported diverse real-world projects, reproduced a rehabilitation pipeline reference-query results, and produced conformant survey QuestionnaireResponses.
- Consistency between profile and implementation: The survey comparison exposed potential differences between profile requirements
- Diagnostic findings: Reviewing validation failures uncovered problems in published profiles and examples, as well as dependency and terminology-resolution issues.
- Explicit authoring requirements: Source-specific mappings still require authored input, and generator limitations remain. Successful transformation does not establish profile conformance.
- Targeted assistance: Agent repair corrected some omitted choice-element mappings, indicating a potential future improvements for feedback-guided completion.

Conclusion

Profile-driven generation provides executable, reviewable FHIR StructureMaps while keeping source-specific decisions explicit, reducing separate implementation of supported structural requirements. Explicit source mappings and unresolved requirements make the remaining authoring decisions reviewable and provide defined targets for future assistance. Publishing and versioning these transformation artifacts alongside profiles could make transformation logic a reusable component of implementation guides and clarify the decisions left to implementers.

References

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¹ <https://aehrc.github.io/redmatch/>

² <https://fume.health>

³ <https://matchbox.health>

Keywords: FHIR, ETL, Reusability, Transformation, FHIR Mapping Language

